

Appendix A - Analysis of the Selected Cases

This appendix brings together a systematization of the selected cases for the research, organizing documentary evidence on digital game development processes through the GQM and Gioia analytical frameworks. Its purpose is to complement the main discussion by making explicit the relationship between the prescribed task, the real engineering activity, and the main frictions observed throughout development. In this way, the material presented makes it possible to trace, comparatively, how governance, communication, and early validation failures become recurring patterns of technical instability and organizational misalignment.

CASE A1 - MOONLIGHTER (DIGITAL SUN)

1. Case identification

- **Project:** Moonlighter.
- **Studio:** Digital Sun.
- **Organizational scale:** studio transitioning from a services model to its own IP.
- **Team composition in the project:** about 6 full-time people.
- **Documentary sources:** technical interview, development account, and public statement by the CEO.

2. Context and engineering activity

Digital Sun originally operated as a services studio and had completed approximately 60 projects for third parties before consolidating Moonlighter as its own IP. The game began as a side project developed alongside outsourcing work, under strong constraints of time, budget, and staff availability. The friction appears in the need to sustain external contracts to finance the studio's infrastructure, which pushed the team toward a logic of progressive improvisation rather than a fully structured pre-production phase. This origin condition is relevant because the project's very architecture had to be built while the studio was still negotiating its organizational transition. Instead of a stabilized prescribed task from the outset, the actual engineering activity was marked by incremental decisions, system integration, and adjustments made under operational pressure.

3. Extraction via the GQM paradigm

Matrix A1.1 - Pre-production governance

Goal	Question	Metric	Finding
Establish solid technical foundations to ensure interoperability between the dungeon and shop management systems.	How did the transition from a services model affect the discipline of systematic prototyping during pre-production?	Development accounts based on improvisation and acknowledgment of “downsides” stemming from the lack of foundations.	The need to maintain cash flow through outsourcing suppressed formal prototyping. The actual activity was marked by system integration without prior validation, generating compliance failures in the final product.
Technical Source URL: https://mcvuk.com/development-news/when-we-made-moonlighter/ Reddit Source URL: https://www.reddit.com/r/IAmA/comments/4qa6nc/iama_the_ceo_of_digital_sun_games_currently/			

Matrix A1.2 - Sustainability of the economic core loop

Goal	Question	Metric	Finding
Synchronize combat and sales mechanics through a balanced and trackable economy.	Did the technical decision to omit a conventional experience system in favor of a gold-based economy result in pacing failures?	Requires 10 to 12 months of post-launch support and multiple balancing patches to correct the progression curve.	The volatile decision-making regarding the progression system created a compliance debt, requiring an exhaustive cycle of belated corrections.
Technical Source URL: https://mcvuk.com/development-news/when-we-made-moonlighter/ Reddit Source URL: https://www.reddit.com/r/IAmA/comments/4qa6nc/iama_the_ceo_of_digital_sun_games_currently/			

Matrix A1.3 - Human Capital Governance

Goal	Question	Metric	Found
Maintaining operational stability through personnel sharing within a multi-brand structure.	How did dividing staff between outsourced projects and in-house IP affect the integrity of the development schedule?	A "painful" learning experience regarding prioritization of efforts and exceeding the original schedule.	The multi-brand governance structure fragmented the team's technical focus. The conflict between financial survival and innovation disrupted the synchronicity between planning and execution, highlighting that weaknesses in personnel governance directly impact the technical sustainability of the product.
Technical Source URL: https://www.gamedeveloper.com/design/launching-a-successful-indie-game-in-2018-the-i-moonlighter-i-story Reddit Source URL:https://www.reddit.com/r/IAmA/comments/4qa6nc/iama_the_ceo_of_digital_sun_games_currently/			

4. Qualitative analysis framework - Gioia Method

Table A1.4 - Qualitative Analysis Structure of Case A1 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The project started very small and has been growing."	Instability in the core design loop	The absence of a stable design goal at the start of the project prevented the crystallization of the initial technical architecture.	Scope Instability
"We did things as we went along."	Suppression of technical validation	The account of improvisation is interpreted as a deliberate abandonment of formal requirements testing processes under deadline pressure.	Pre-Production Failure

"We shared staff between four companies."	Fragmentation of human resources	The division of the technical team among different brands and business objectives led to a dilution of knowledge about the product.	Fragmentation and Misalignment
"It was difficult to learn where to prioritize the staff."	Organizational misalignment	Uncertainty in leadership reveals a failure in strategic governance that prevents the maintenance of a common knowledge base.	Fragmentation and Misalignment
"Downsides in the game resulting from a lack of foundation"	Systemic degradation due to inertia	Admission of permanent product flaws due to shortcut decisions made in pre-production.	Technical Debt
"We need 12 months of post-launch patches."	Accumulation of compliance debt	The exhaustive post-launch correction time is the operational cost generated by the lack of traceability of requirements during production.	Technical Debt
"Gold as a core loop... without an experience system"	Volatility of the economic model	The reliance on a single metric broke the traceability between design and actual behavior.	Scope Instability

5. Final considerations of the case

The fragility of compliance in Moonlighter did not stem from technical incompetence, but from a failure of strategic governance. The transition from a service model to proprietary IP created a space of low prototyping formalization, in which the studio's need for survival superseded the prescribed task of rigorous engineering. The case suggests that fragmentation and misalignment of focus acted as central drivers of the accumulation of technical debt, making software stability dependent on prolonged maintenance after launch.

CASE A2 - THE WITCHER 3: WILD HUNT (CD PROJEKT RED)

1. Case identification

- **Project:** The Witcher 3: Wild Hunt.
- **Studio:**CD Projekt RED.
- **Organizational scale:**Expansion from 90 to 250 developers during production.
- **Target platforms:**Three next-generation platforms.
- **Documentary sources:**Technical presentation, expert interview, public discussion, and user testimonial.

2. Engineering context and activity

CD Projekt RED faced the challenge of tripling its human capital to simultaneously deliver a title across three next-generation platforms. The engineering activity, understood as the effective result of execution under pressures of scale and uncertainty, revealed the unfeasibility of manual processes for asset governance in a large-scale open world. The need to replace human oversight with algorithmic automation, in order to preserve the system's performance budget, highlights the Ontological Friction in this case.

This condition manifests itself on multiple fronts: visibility, aesthetic fidelity, sensory immersion, and portability. In all of them, the prescribed task formalized in pre-production clashed with the physical limitations of the engine and the scale of creative ambition.

3. Extraction via GQM paradigm

Matrix A2.1 - Scale and Visibility Governance

Goal	Question	Metric	Analysis/Response
To efficiently render only visible objects in order to maintain frame rate stability at a massive scale.	How can occlusion be automated when manual marking is physically impossible in an open world?	Implementation of an automated occlusion culling system via Umbra 3 middleware.	The massive scale of the artifact broke the viability of manual tasks; governance shifted to algorithmic automation as the only way to avoid systemic performance collapse and ensure technical compliance across platforms.
Technical Source URL: https://www.gdcvault.com/play/1022005/Solving-Visibility-and-Streaming-Problems Reddit Source URL: https://www.reddit.com/r/witcher/comments/2zpx24/the_witcher_3_wild_hunt_umbra_3_gdc_presentation/			

Matrix A2.2 - Aesthetic Fidelity Optimization

Goal	Question	Metric	Analysis/Response
Optimize the rendering subsystem to operate in a scalable manner across different hardware.	What technical adjustments mitigate the excessive cost of complex shaders without sacrificing visual quality?	Reports of "extremely expensive" shaders for the GPU and the replacement of per-pixel calculations with per-vertex calculations in the fog.	The initial aesthetic ambition created a processing bottleneck that required mathematical engineering solutions to enable execution within the physical limitations of the consoles.
Technical Source URL: https://80.lv/articles/the-witcher-3-interview-a-deeper-look-into-redengine-3/ Reddit Source URL: https://www.reddit.com/r/witcher/comments/36fcl5/hairworks_is_destroying_my_fps/			

Matrix A2.3 - Sensory Immersion Compliance

Goal	Question	Metric	Analysis/Response
To promote the player's connection with the virtual environment through audiovisual harmony.	Do technical limitations in audio and lighting integration hinder spatial perception and user immersion?	User reports indicate "studio-sounding" dialogue outdoors and inconsistent lighting in cutscenes.	The failure in traceability between audio capture and the physics of the virtual environment generated sensory dissonance, revealing governance flaws in the integration of high-fidelity assets between audio and world engineering.
Technical Source URL: https://rsdjournal.org/index.php/rsd/article/view/43328 Reddit Source URL: https://www.reddit.com/r/witcher/comments/3792m8/the_witcher_3_speech_audio_everything_sounds_like/			

4. Qualitative analysis framework - Gioia Method

Table A2.4 - Qualitative Analysis Structure of Case A2 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Manually marking occluders is impossible at this size."	Elimination of manual processes	The massive scale prevents direct human control, forcing a shift to algorithmic automation.	Fragmentation and Misalignment
"The hair shader was initially extremely expensive for the GPU."	Dissonance between ambition and infrastructure.	The mismatch between aesthetic requirements and physical limitations reveals a failure to anticipate constraints during pre-production.	Pre-Production Failure
"The audio sounds like it's in a studio... it breaks the immersion."	Fragmentation of sensory fidelity	The lack of traceability between the sound asset and the physics of the setting indicates silos between audio and engineering.	Fragmentation and Misalignment
"We had to do a lot of tricks to make it run better."	Accumulation of palliative solutions	The need for last-minute fixes is evidence of technical debt generated by under-planning.	Technical Debt
"Lighting during cutscenes noticeably brighter"	Inconsistency in visual traceability.	The failure to maintain visual standards across software states indicates a breakdown in asset governance.	Technical Debt
"Tripling the team... a mix of all cultures and inspirations"	Organizational misalignment by scale	Rapid team growth without strict onboarding protocols dilutes the trail of technical decisions.	Fragmentation and Misalignment
"First game released simultaneously on 3 platforms"	Complexity of cross-portability	Simultaneous multi-platform ambition without a stable baseline generates reactive compliance liabilities.	Technical Debt

5. Final considerations of the case

The analysis of The Witcher 3 demonstrates that technical stability in AAA projects depends on a delicate balance between massive automation and the acceptance of sensory dissonance. The trail of evidence shows that Ontological Friction emerges from the tension between extreme realism and the inability of the physical infrastructure to process such ambition without late engineering "tricks." The

case suggests that Fragmentation and Misalignment between development silos and software engine limitations constitute one of the main generators of Technical Debt.

CASE A3 - BRÜTAL LEGEND (DOUBLE FINE)

1. Case identification

- **Project:** Brutal Legend.
- **Studio:** Double Fine.
- **Organizational scale:** Medium-sized studio experiencing significant publishing instability.
- **Central condition of the case:** Chaotic transition between editors and legal pressure during the final cycle.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

Double Fine faced an unstable transition of publishers, from Sierra to Activision and subsequently Electronic Arts, during a period also marked by legal disputes. Engineering activity, understood as the effective result of execution under external governance pressures, revealed that technical and executive leadership was diverted from its core function to handle legal tasks during the Alpha and Gold phases. We observed Ontological Friction in the difficulty of the infrastructure simultaneously supporting an imposed genre pivot and an unplanned avalanche of content at the end of the cycle.

This unstable governance condition compromised the continuity between planning, execution, and final stabilization of the artifact. Instead of a controlled production flow, the project began to operate under decisional interruptions, resource redirection, and last-minute reactive corrections.

3. Extraction via GQM paradigm

Matrix A3.1 - Scope and Gender Governance

Goal	Question	Metric	Analysis/Response
To integrate the Action-Adventure and RTS genres in a cohesive and transparent way for the user.	How has the design pivot imposed by publishers affected the perception of technical quality and fluidity of the software?	Perception of "horrible gameplay" in the core systems and description of the product as an "exercise in mediocrity".	The late inclusion of action elements in a project originally structured as an RTS (Real-Time Strategy) generated architectural inconsistency. Engineering failed to stabilize traceability between the requirements of both types of systems, resulting in systems perceived as strange and lacking unification.

Technical Source URL:

<https://www.gamedeveloper.com/game-platforms/postmortem-behind-the-scenes-of-i-brutal-legend-i->

Reddit Source URL:

https://www.reddit.com/r/patientgamers/comments/n9zb3a/brutal_legend_brutal_slog/

Matrix A3.2 - Infrastructure Scalability

Goal	Question	Metric	Analysis/Response
Maintaining the stability of the build and version control infrastructure in the face of a massive increase in assets.	Did the failure in the data infrastructure compromise the ability to refine complex game systems?	A jump from 2.5 GB to 9 GB of data in four months, resulting in cascading effects and bottlenecks on the server.	Underestimating the simultaneous delivery of assets created a technical liability, diverting engineers from their original tasks to containment activities and preventing the optimization of critical systems.

Technical Source URL:

<https://www.gamedeveloper.com/game-platforms/postmortem-behind-the-scenes-of-i-brutal-legend-i->

Reddit Source URL:

https://www.reddit.com/r/patientgamers/comments/n9zb3a/brutal_legend_brutal_slog/

Matrix A3.3 - Leadership Stabilization

Goal	Question	Metric	Analysis/Response
Ensure the software is polished and technically compliant during the Alpha/Gold milestone.	What is the impact of disrupting technical leadership decision-making processes on the ultimate integrity of the user experience?	Account of leaders diverted to reading court documents and testimonies during the Alpha period.	The loss of human governance over the trail of technical evidence at the end of the cycle made it difficult to correct immersion and balancing errors, reducing the effectiveness of final product oversight.

Technical Source URL:

<https://www.gamedeveloper.com/game-platforms/postmortem-behind-the-scenes-of-i-brutal-legend-i->

Reddit Source URL:

https://www.reddit.com/r/patientgamers/comments/n9zb3a/brutal_legenda_brutal_slog/

4. Qualitative analysis framework - Gioia Method

Table A3.4 - Qualitative Analysis Structure of Case A3 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Critics confused by the sudden shift to RTS"	Volatility of gender requirements	The change in the technical premise at an advanced stage broke the design baseline, creating an unstable hybrid artifact.	Scope Instability
"Leadership diverted to reading court documents"	Fragmentation of leadership focus	The disruption of human governance has created decision-making silos, hindering technical stabilization.	Fragmentation and Misalignment
"We went from 2.5 GB to 9 GB of data in weeks."	Data infrastructure bottleneck	The inability to absorb the volume of content indicates a failure to anticipate physical limitations.	Technical Debt
"Engineers drawn into firefighting"	Accumulation of compliance debt	Reactive correction time reflects the cost of a lack of traceability between assets and execution.	Technical Debt
"We underestimated the avalanche of final content."	Suppression of predictive planning	The absence of a scalability model in pre-production forced stopgap solutions.	Pre-Production Failure
"The demo didn't show what the game was really like."	Marketing traceability disruption	The mismatch between the product promise and the actual gameplay reveals a failure in requirements governance.	Scope Instability

"Every aspect of the gameplay ended up being weak."	Degradation of systemic integrity	The attempt to cover multiple genres without dedicated infrastructure resulted in a loss of quality.	Technical Debt
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5. Final considerations of the case

Brütal Legend suggests that fragility in external governance was a central factor in the loss of technical compliance. The trail of evidence shows that, even with agile practices, technical stability deteriorates when governance is captured by extra-technical factors and infrastructure is not proactively scaled for the mass production phase. The case indicates that fragmentation and misalignment can emerge as a state produced by decision silos imposed by administrative crises.

CASE A4 - BORDERLANDS (GEARBOX SOFTWARE)

1. Case identification

- **Project:** Borderlands.
- **Studio:** Gearbox Software.
- **Organizational scale:** Team experiencing rapid growth during production.
- **Central condition of the case:** Documentary obsolescence, a shift in aesthetic direction, and a centralization of critical knowledge.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

Gearbox began developing Borderlands with a small team and high-level goals, using a blue sky thinking approach. The prescribed task for requirements communication and governance was the use of a centralized wiki. In actual engineering activity, however, the design evolved faster than the documentation was updated, making the wiki obsolete and the record-keeping discipline inconsistent. This led to the erosion of technical memory: the team became unable to recall the reasons behind previous architectural decisions, resulting in unproductive meetings and features that were not easily visible to management.

This dynamic shows that the project's complexity stemmed not only from the scale of the content, but also from the inability of the documentary infrastructure to keep pace with the execution. Instead of consolidating a shared decision repository, the process began operating with outdated records, ambiguous criteria, and an increasing dependence on individual tacit knowledge.

3. Extraction via GQM paradigm

Matrix A4.1 - Documentation Governance and Traceability

Goal	Question	Metric	Analysis/Response
Maintain visibility and control over the design status during team scaling.	How have the obsolescence of prescribed documentation and the lack of discipline in updating it affected the traceability of engineering decisions?	Occurrence of meetings without decision-making and implementation of undocumented functionalities.	The scalability failure of the prescribed tool broke technical traceability. Engineering activity operated in a documentation vacuum, resulting in analysis paralysis and loss of technical memory regarding the reasons for the architectural choices.
Technical Source URL: https://www.gamedeveloper.com/game-platforms/postmortem-highlights-behind-the-scenes-of-i-b-orderlands-i- . Reddit Source URL: https://www.reddit.com/r/Borderlands4/comments/1fdkd12/so_are_we_going_to_do_anythin-g_about_the_wiki/ .			

Matrix A4.2 - Governance of Artistic Assets

Goal	Question	Metric	Analysis/Response
Establish clear processes for the creation and evaluation of technical art assets.	Has the lack of a technical definition of "finished" compromised the governance of artistic production?	Degree of uncertainty among art leaders regarding the final state of assets before a radical style change.	The absence of an objective technical baseline for finalizing assets in the realistic style led to massive rework. Compliance was only stabilized with the externalization of clear criteria after the change to the visual inked style.
Technical Source URL: https://www.gamedeveloper.com/game-platforms/postmortem-highlights-behind-the-scenes-of-i-b-orderlands-i- . Reddit Source URL: https://www.reddit.com/r/Borderlands2/comments/1o3la4q/help_please/ .			

Matrix A4.3 - Knowledge Management and Expertise

Goal	Question	Metric	Analysis/Response
Develop a scalable and easily debuggable mission system.	How did the lack of refactoring of the custom scripting system affect the productivity of the level design team?	Ratio of specialists per system (1:1) and increased workload for a single individual.	Prioritizing delivery over architecture created a knowledge silo. The technical inability to refactor the system forced reliance on the tacit rationality of a single developer, hindering maintenance.
Technical Source URL: https://www.gamedeveloper.com/game-platforms/postmortem-highlights-behind-the-scenes-of-i-b-orderlands-i-. Reddit Source URL: https://www.reddit.com/r/Borderlands2/comments/1o3la4q/help_please/.			

4. Qualitative analysis framework - Gioia Method

Table A4.4 - Qualitative Analysis Structure of Case A4 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Designs evolved faster than wiki pages"	Obsolescence of prescribed documentation	The speed of technical execution exceeded the formal record, highlighting the mismatch between pace and infrastructure.	Fragmentation and Misalignment
"Difficulty in remembering why certain decisions were made"	Erosion of technical memory	The loss of evidence about the past generates decisional inertia and analytical paralysis.	Fragmentation and Misalignment
"It was uncertain whether the assets were ready in a realistic style."	Ambiguity in delivery criteria	The lack of an objective technical baseline for completion reveals a failure in defining verifiable requirements.	Pre-Production Failure
"The system was never refactored to be user-friendly."	Suppression of architectural refactoring	The decision to prioritize the release over the sustainability of the code generates technical rigidity.	Scope Instability

"Only one person understood the system inside and out."	Centralization of critical knowledge	The reliance on a single individual for debugging embodies the operational risk of technical silos.	Technical Debt
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5. Final considerations of the case

Borderlands demonstrates that ultimate technical stability was achieved through a radical shift in artistic direction, which provided the necessary externalization for clearer technical criteria. However, the project's foundation suffered from fragmentation and misalignment caused by poorly scalable documentation tools and knowledge silos. This case reinforces that pre-production failure and the accumulation of technical debt are systemic patterns independent of the game's creative nature.

CASE A5 - BIOSHOCK (2K BOSTON / 2K AUSTRALIA)

1. Case identification

- **Project:** BioShock.
- **Studio:** 2K Boston / 2K Australia, originally Irrational Games.
- **Organizational scale:** Multidisciplinary development with a transition in positioning and increased complexity.
- **Central condition of the case:** Late repositioning, system rework, and weak technical traceability.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

The Irrational Games team began the project with the prescribed task of developing a hybrid between RPG and FPS, continuing the legacy of System Shock 2. However, the actual engineering activity, understood as the effective result of execution under the pressures of multidisciplinary development and design uncertainty, showed that the absence of a sufficiently clear product message led the team to invest considerable time in a direction that would not solidify as the final form of the game. We observed a type of Ontological Attrition where late scope corrections were necessary to convert more complex systems into more accessible interactions, compatible with the experience of a linear shooter.

3. Extraction via GQM paradigm

Matrix A5.1 - Strategic Software Positioning

Goal	Question	Metric	Analysis/Response
To develop commercially viable software as a competitive first-person shooter.	How did the initial lack of clarity regarding the technical scope, between RPG and FPS, impact the usefulness of the implemented systems?	The need to simplify deep systems and extend the timeline following the delayed repositioning.	The lack of strategic clarity hindered the early stabilization of the software architecture. This was interpreted as evidence of a lack of requirements, requiring the team to perform significant refactoring to remove complexities that did not align with the actual expected activity of the end user.
Technical Source URL: https://www.gamedeveloper.com/view/feature/3774/postmortem_2k_boston2k_php. Reddit Source URL: https://www.reddit.com/r/patientgamers/comments/sntrzl/first_time_finishing_bioshock_2007_the_broken/.			

Matrix A5.2 - Production flow efficiency

Goal	Question	Metric	Analysis/Response
Optimize technical productivity and reduce development iteration cycles.	Did the absence of code governance standards and slow infrastructure processes compromise engineering efficiency?	Time spent searching for script actions without naming conventions and hours consumed in slow baking processes.	The absence of technical baselines and naming conventions compromised internal traceability. Operational costs were interpreted as a manifestation of fragmentation, with a direct impact on team productivity and production infrastructure.

Technical Source URL:

https://www.gamedeveloper.com/view/feature/3774/postmortem_2k_boston2k_.php.

Reddit Source URL:

https://www.reddit.com/r/Bioshock/comments/1qayvoi/skyline_test_footage_from_2008_early_sky_box_and/.

Matrix A5.3 - Evidence-based governance

Goal	Question	Metric	Analysis/Response
Implement balance adjustments based on real gameplay metrics.	Was the technical log system capable of providing actionable metrics for decision-making ?	Inadequacy of metric analysis results and reliance on assumptions based on testers using shortcuts.	The failure in the data collection infrastructure compromised the externalization of knowledge about player behavior. Without auditable evidence, engineering became dependent on less reliable assumptions, which produced a mismatch between design intent and execution.
Technical Source URL: https://www.gamedeveloper.com/view/feature/3774/postmortem_2k_boston2k_.php . Reddit Source URL: https://www.reddit.com/r/patientgamers/comments/sntrzl/first_time_finishing_bioshock_2007_the_broken/ .			

4. Qualitative analysis framework - Gioia Method**Table A5.4 - Qualitative Analysis Structure of Case A5 (Gioia Method)**

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
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"We spend most of our time playing the wrong game."	Volatility of strategic positioning	The absence of a stable goal hindered the sustainable consolidation of the software architecture.	Scope Instability
"No convention for how to name script actions"	Lack of technical baselines	The failure to establish governance standards reduced productivity and made maintenance more difficult.	Pre-Production Failure
"The logging system was inadequate and the results useless."	Erosion of performance traceability	The inability to generate auditable evidence hindered the alignment between design and physical capacity.	Fragmentation and Misalignment
"Vertical slice was unnavigable and was discarded."	Architectural validation failure	The difficulty in translating the creative vision into a functional prototype reveals an early mismatch.	Pre-Production Failure
"We had to simplify systems so the user could understand them."	Accumulation of compliance debt	The loss of technical depth due to insufficient polishing time was a direct cost of poor traceability.	Technical Debt

5. Final considerations of the case

BioShock indicates that the project's technical liabilities were primarily fueled by a failure of strategic governance, materialized in the late repositioning of the genre. The case suggests that fragmentation and misalignment emerge when technical tools, such as scripts and logs, fail to keep pace with creative ambition, forcing course corrections that reduce the artifact's original conformity and increase rework on the production infrastructure.

CASE A6 - MASS EFFECT 2 (BIOWARE)

1. Case identification

- **Project:** Mass Effect 2.
- **Studio:** BioWare.
- **Organizational scale:**Producing a sequel in an established franchise, under high pressure to maintain continuity.
- **Central condition of the case:**Governance restructuring, complex data persistence, and erosion of technical memory.

- **Documentary sources:** Technical analysis, expert interview, and associated public discussion.

2. Engineering context and activity

BioWare faced the challenge of correcting perceived weaknesses in the first Mass Effect within a relatively short development cycle, while maintaining the expectation of continuity for an already established franchise. The actual engineering activity, understood as the effective result of execution under the pressures of continuity and audience response, revealed a significant shift in project governance: the studio stopped operating in isolated silos and began organizing work in interdisciplinary pods, bringing together writers, technical designers, and artists. This case highlights a type of Ontological Friction linked to the complexity of managing multiple persistence variables inherited from previous software states, especially in the handling of save games and narrative continuity.

3. Extraction via GQM paradigm

Matrix A6.1 - System Architecture Optimization

Goal	Question	Metric	Analysis/Response
Reduce excessive micromanagement and inefficiencies inherited from the previous game by streamlining systems.	Did the technical simplification of the systems preserve the player's engagement with the requirements of choice and consequence?	Removal of independent systems and automation of class paths.	The suppression of systems deemed incomplete in the previous design was codified as a governance strategy geared towards combat efficiency. The case suggests that this decision reduced the complexity of the RPG simulation while preserving greater clarity for the core player experience.
Technical Source URL: https://www.gamedeveloper.com/game-platforms/analysis-i-mass-effect-2-i-s-surprising-genre-experiment . Reddit Source URL: https://www.reddit.com/r/masseffect/comments/lj1urb6/a_summary_of_mass_effect_2s_main_plot/ .			

Matrix A6.2 - Data Persistence Traceability

Goal	Question	Metric	Analysis/Response
To create a continuous narrative in which the player's decisions coherently alter the software architecture.	Did implementing hundreds of persistence variables prevent logical dissonance in user interaction?	Volume of indexed variables and reports of inconsistent NPC behavior.	Although the infrastructure was able to process a large volume of variables, the case reveals weaknesses in logical governance. Data traceability was maintained at a technical level, but the correspondence between narrative variables and observed behavior showed disruptions at specific points in the experiment.
Technical Source URL: https://www.pcworld.com/article/524447/mass_effect_2_interview_p1.html. Reddit Source URL: https://www.reddit.com/r/masseffect/comments/ljlurb6/a_summary_of_mass_effect_2s_main_plot/.			

Matrix A6.3 - Human capital and knowledge governance

Goal	Question	Metric	Analysis/Response
Maintaining ludonarrative synergy through veteran retention and the integration of new talent.	How have leadership turnover and the arrival of new developers affected the integrity of the narrative architecture?	Reports of blind spots in new talent and documented departures of senior managers.	The loss of experienced personnel was coded as a process of erosion of technical memory. Without the tacit knowledge accumulated by veterans, new team members began to operate with gaps in the system's motivational logic, which affected the consistency of the narrative experience.
Technical Source URL: https://www.vice.com/en/article/why-mass-effect-2-was-the-series-pinnacle-and-what-to-learn-from-it/.			

Reddit Source URL:

https://www.reddit.com/r/masseffect/comments/1j1urb6/a_summary_of_mass_effect_2s_main_plot/

4. Qualitative analysis framework - Gioia Method

Table A6.4 - Qualitative Analysis Structure of Case A6 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"We work in pods with writers and programmers together."	Interdisciplinary integration through proximity	The shift from silos to pods was a governance response to maintain traceability between narrative intent and engine.	Fragmentation and Misalignment
"Talented new professionals will have more blind spots"	Erosion of technical memory due to turnover.	The loss of skilled human capital results in the loss of undocumented subjective requirements.	Fragmentation and Misalignment
"The save game literally contains hundreds of things."	Trans-software persistence complexity	Managing legacy states creates a massive technical liability that requires constant validation.	Technical Debt
"NPCs reprimanding Shepard right after he saved them"	Narrative conformity dissonance	The failure to translate technical variables into coherent behaviors reveals a disconnect between data and experience.	Technical Debt
"We wanted to ensure that every problem was addressed."	Reactive requirements optimization	The focus on correcting the predecessor's mistakes overshadowed some of the original design's depth.	Scope Instability
"Remove planetary exploration on the Mako vehicle"	Strategic scope truncation	The elimination of systems deemed incomplete was a simplification aimed at focusing on squad-based combat.	Scope Instability
"Writers plotted stories like D&D modules"	Narrative synchronicity versus technical feasibility	The mismatch between the creative mental model and the game engine reveals a pre-production flaw.	Pre-Production Failure

5. Final considerations of the case

Mass Effect 2 reveals a governance paradox: although the studio implemented an infrastructure capable of tracking hundreds of variables, the user experience was still marked by logical dissonances at specific points. This suggests that the fragmentation and misalignment between narrative logic and technical variables, coupled with the technical debt accumulated from the legacy of previous software states, makes visible the fragility of the decision-making process in the transition between episodes of a complex franchise.

CASE A7 - INKBOUND (SHINY SHOE)

1. Case identification

- **Project:** Inkbound.
- **Studio:** Shiny Shoe.
- **Organizational scale:** A studio already experienced in roguelikes, transitioning to online co-op.
- **Central condition of the case:** Gender pivot, monetization crisis, and localization fragility.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

Following the success of Monster Train, Shiny Shoe sought to expand its roguelike expertise into an online cooperative environment. The actual engineering work, understood as the effective result of execution under market pressures and uncertainty, showed that the prescribed task of maintaining a strictly card-based game was technically inadequate for cooperative play, especially due to the game's pacing. The need to preserve fluidity and usability led to a structural pivot to a hybrid real-time and turn-based system, as well as a revision of the business model after negative community reaction to monetization in Early Access.

3. Extraction via GQM paradigm

Matrix A7.1 - Gender governance and pace

Goal	Question	Metric	Analysis/Response
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Expand the roguelike strategy mechanics to online cooperative multiplayer.	Did the initial prototype, based exclusively on cards, meet the requirements for rhythm and fluidity for multiple simultaneous players?	Evaluation of the game pace in the initial prototypes and the stated need to pivot to a hybrid system.	The mechanics inherited from previous projects proved incompatible with the new multiplayer infrastructure. Governance failed to validate group gameplay during pre-production, leading to the need to alter the software's basic premise to ensure fluidity.
Technical Source URL: https://www.gamedeveloper.com/production/postmortem-inkbound-s-journey-in-early-access. Reddit Source URL: https://www.reddit.com/r/roguelites/comments/1g6co6m/inkbound_from_the_creators_of_monster_train_and/.			

Matrix A7.2 - Strategic Monetization Governance

Goal	Question	Metric	Analysis/Response
To implement a business model capable of sustaining continuous updates without segregating the community.	Was the decision to include in-game monetization in Early Access accepted as a legitimate governance practice?	Volume of negative feedback and complete removal of the monetization system during development.	There was a disconnect between the business model and user sentiment. The failure to anticipate the commercial and ethical impact of monetization generated significant compliance debt, requiring refactoring of economic systems already in operation.

Technical Source URL:

<https://www.consolecreatures.com/inkbound-is-removing-all-battle-pass-and-in-game-monetization-in-latest-update/>.

Reddit Source URL:

https://www.reddit.com/r/Games/comments/177190v/inkbound_is_removing_its_ingame_monetization/.

Matrix A7.3 - Traceability of location requirements

Goal	Question	Metric	Analysis/Response
Maximize the ability to iterate rapidly in design without relying on slow translation flows.	What was the impact of releasing the software only in English on business stability and user retention?	Refund rates specifically driven by the lack of Chinese language support in Early Access.	Prioritizing technical agility over global linguistic requirements resulted in significant commercial losses. The lack of Chinese language support was interpreted as eroding the traceability of the artifact's success, directly impacting its financial sustainability.

Technical Source URL:

<https://www.gamedeveloper.com/production/postmortem-inkbound-s-journey-in-early-access>.

Reddit Source URL:

https://www.reddit.com/r/InkBound/comments/1b2iu3a/how_is_the_single_player_experience_does_the/.

4. Qualitative analysis framework - Gioia Method**Table A7.4 - Qualitative Analysis Structure of Case A7 (Gioia Method)**

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The pace was too slow... we pivoted to a hybrid."	Radical gender pivot	The mismatch between the legacy mechanics and the multiplayer infrastructure required a change in the basic premise.	Scope Instability
"We were taken by surprise by the amount of negative feedback."	Fragility of market predictability	The failure to anticipate the reaction to monetization systems reveals a blind spot in requirements governance.	Fragmentation and Misalignment
"Suggestions for solo vs. co-op were at odds."	Conflict of gameplay requirements	The fragmentation of focus across user profiles prevented a unified technical overview.	Fragmentation and Misalignment
"Release only in English to iterate quickly"	Suppression of localization requirements	The decision to ignore language markets in order to accelerate the code cycle was a planning failure.	Pre-Production Failure
"We were forced to completely remove monetization."	Accumulation of business compliance debt	The need to refactor the revenue model after launch is a direct cost of the lack of traceability between design and user.	Technical Debt
"First major online game... centralized hosting"	Lack of experience in network infrastructure	The adoption of a new architecture without a prior basis increased the systemic risk that was not anticipated in the planning.	Pre-Production Failure

"Refund principal: Chinese language not supported"	Erosion of traceability of global success	The failure to track regional requirements resulted in a loss of market share that will be difficult to recover.	Technical Debt
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5. Final considerations of the case

Inkbound suggests that fragmentation and misalignment between audience expectations and governance strategy, especially in monetization and localization, were central drivers of the fragility of the decision-making process. The case indicates that, even with feedback tools, technical and commercial stability breaks down when governance suppresses fundamental requirements in favor of poorly dimensioned iteration agility.

CASE A8 - DRAGON AGE II (BIOWARE)

1. Case identification

- **Project:** Dragon Age II.
- **Studio:** BioWare.
- **Organizational scale:** Development under high pressure, with the team simultaneously distributed across multiple AAA projects.
- **Central condition of the case:** Massive asset recycling, systemic simplification, and reduced space ambition.
- **Documentary sources:** technical analysis and associated public discussion.

2. Engineering context and activity

BioWare faced the challenge of developing Dragon Age II in a significantly reduced timeline, while the studio's human capital was overstretched by the simultaneous development of Mass Effect 3 and Star Wars: The Old Republic. The engineering activity, understood as the effective result of executing a task under operational pressures and time constraints, revealed a survival strategy based on the massive recycling of assets and the radical simplification of systems. Ontological friction becomes visible in the delivery of a small, unchanging, and shallow virtual world, which diverged head-on from the prescribed task of expanding the epic begun in Origins.

3. Extraction via GQM paradigm

Matrix A8.1 - Scope and Asset Governance

Goal	Question	Metric	Analysis/Response
To offer a sequel with the same scale and variety of settings as its predecessor.	How has the division of efforts across multiple AAA projects affected the integrity of the game world infrastructure?	Systematic occurrence of recycled scenarios and identical dungeon layouts with paths only visually blocked.	The studio opted for compulsive asset recycling to meet the schedule, resulting in technical infrastructure debt that degraded the perceived technical quality and user immersion.
Technical Source URL: https://www.popmatters.com/149535-dragon-age-origins-vs-dragon-age-ii-2495901174.html (Ref:). Reddit Source URL: https://www.reddit.com/r/dragonage/comments/3bbdv8/da2_spoilers_why_did_da2_have_repeating/ (Ref:).			

Matrix A8.2 - Narrative Conformity and Design

Goal	Question	Metric	Analysis/Response
Evolve the choice and consequence structure inherited from the franchise.	Did the shift to a linear narrative focused on the protagonist Hawke impact the traceability of the player's decisions?	Accounts of choices without significant consequence and the perception that the protagonist is merely a bystander in systemic changes.	There was a failure in externalizing the narrative complexity. The transition to a disjointed structure prevented the player's choices from dynamically altering the software architecture, highlighting a pre-production failure to maintain the traceability rigor of Origins.

Technical Source URL:

<https://www.popmatters.com/149535-dragon-age-origins-vs-dragon-age-ii-2495901174.html> (Ref:).

Reddit Source URL:

https://www.reddit.com/r/dragonage/comments/13dkkx7/how_important_are_choices_and_consequence_to_you/ (Ref:).

Matrix A8.3 - Sustainability of the Experience

Goal	Question	Metric	Analysis/Response
Maintain engagement through balanced combat and progression systems.	Did simplifying the mechanics result in a technically satisfactory experience or in systemic monotony?	Criticisms included the monotonous combat and design choices deemed lazy due to reduced polishing time.	The suppression of systemic complexity in favor of production agility has broken the traceability between the promise of RPG depth and actual combat activity, consolidating a state of Scope Instability.
Technical Source URL: https://www.popmatters.com/149535-dragon-age-origins-vs-dragon-age-ii-2495901174.html (Ref:). Reddit Source URL: https://www.reddit.com/r/dragonage/comments/39ptj0/spoilers_all_dragon_age_and_story_structure/ (Ref:).			

4. Qualitative analysis framework - Gioia Method**Table A8.4 - Qualitative Analysis Structure of Case A8 (Gioia Method)**

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
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"Blatantly identical environments... dungeons recycled a dozen times"	Compulsive asset recycling	The inability to produce new assets forces the reuse of layouts, breaking the integrity of space exploration.	Technical Debt
"Bioware overstretched during the development of ME3 and Old Republic"	Fragmentation of organizational focus	Dividing human capital among multiple AAA projects dilutes the technical oversight necessary for compliance.	Fragmentation and Misalignment
"The plot is a disorganized mess... deeply personal, but disjointed."	Fragmentation of the narrative structure	The loss of a cohesive architectural vision results in a narrative that fails to integrate branching technical requirements.	Scope Instability
"Simplified mechanics mixed with lazy design choices"	Suppression of systemic complexity	Radical simplification is interpreted as a deliberate renunciation of engineering depth in order to save on production cycles.	Pre-Production Failure
"They replaced the grand world with a small, unchanging, and shallow one."	World scalability regression	The decision to restrict the geographic environment is a technical response to the lack of infrastructure to support an open world.	Scope Instability

5. Final considerations of the case

Dragon Age II confirms that technical instability and loss of compliance were not isolated errors, but the direct result of a strategic governance failure in managing the studio's production capacity. The trail of evidence shows that fragmentation and misalignment, caused by the team's dispersal across three simultaneous AAA titles, acted as a trigger for the accumulation of technical debt, manifested in the compulsive recycling of assets and systemic monotony. The case suggests that software stability depends on integrated requirements governance that is compatible with the physical limitations of the production infrastructure.

CASE A9 - MIDDLE-EARTH: SHADOW OF MORDOR (MONOLITH PRODUCTIONS)

1. Case identification

- **Project:** Middle-earth: Shadow of Mordor.
- **Studio:** Monolith Productions.
- **Organizational scale:** Building a new team for a highly ambitious project.
- **Central condition of the case:** Unified organizational focus, complex procedural system, and late feature cuts.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

Monolith Productions faced the challenge of building a new team for an ambitious project, in which many members were working together for the first time. The engineering activity revealed a strategic shift in governance: the studio abandoned the development of multiple titles in parallel to concentrate all efforts on a single game. This focus allowed engineering to optimize runtime systems and tools specifically for the requirements of the Nemesis System, a procedural engine in which each enemy possesses unique personality and memory attributes.

Ontological friction manifests itself in the loss of customers caused by failures in user experience (UX) or constant rework in engineering and development projects, and in the technical impact of late feature cuts, producing what leadership has described as a "double dose of pain".

3. Extraction via GQM paradigm

Matrix A9.1 - Governance of the procedural system

Goal	Question	Metric	Analysis/Response
Implement an innovative AI system that creates narrative and emotional persistence between the player and the enemies.	How did a unified organizational focus and interdisciplinary collaboration enable the conformity of a high-fidelity procedural system?	Reports of complete integration between writing, voice-over, AI, animation, and level design in the Nemesis System.	The convergence of focus allowed engineering to break down traditional silos. The compliance of the Nemesis system was achieved not only through algorithms, but through governance that fostered alignment between art assets and AI logic from the outset.

Technical Source URL:

<https://www.gamedeveloper.com/design/postmortem-monolith-productions-i-middle-earth-shadow-of-mordor-i->

Reddit Source URL:

https://www.reddit.com/r/shadowofmordor/comments/2it9y0/monolith_productions_ama/

Matrix A9.2 - Scope refactoring efficiency

Goal	Question	Metric	Analysis/Response
Manage feature cuts without compromising stability or overloading production.	What is the technical impact and operational cost of replacing cut requirements with new content in the final phase of the cycle?	The pain occurs in two doses: loss of work already done and the creation of a new workload that exceeds what was planned.	The reduction in functionality generated a reactive compliance debt. The need to fill the gap left by the removed requirements demanded a greater engineering effort than the original implementation, revealing weaknesses in the prior scope validation.

Technical Source URL:

<https://www.gamedeveloper.com/design/postmortem-monolith-productions-i-middle-earth-shadow-of-mordor-i->

Reddit Source URL:

https://www.reddit.com/r/shadowofmordor/comments/2it9y0/monolith_productions_ama/

Matrix A9.3 - Infrastructure stability and iteration

Goal	Question	Metric	Analysis/Response
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To provide technical resilience through continuous integration systems and stable builds.	How has investing in customized infrastructure mitigated the risks of fluctuating requirements?	The use of a robust build infrastructure enabled rapid, real-time iteration.	The infrastructure acted as a physical traceability protocol. The build system allowed for the immediate validation of technical goals against real-world gameplay activity, mitigating the fragmentation that often occurs in complex AI projects.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-monolith-productions-i-middle-earth-shadow-of-mordor-i- Reddit Source URL: https://www.reddit.com/r/shadowofmordor/comments/2it9y0/monolith_productions_ama/			

4. Qualitative analysis framework - Gioia Method

Table A9.4 - Qualitative Analysis Structure of Case A9 (Joy Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The entire studio focused on just one game instead of multiple games running in parallel."	Convergence of organizational focus	The centralization of human capital was the governance response to mitigate fragmentation and ensure technical oversight of complex systems.	Fragmentation and Misalignment (Mitigation)
"Cutting jobs created double the pain: job loss and the creation of more jobs."	Reactive scope refactoring	Removing requirements created a technical liability where the effort to redesign the system exceeded the cost of the original maintenance.	Scope Instability

"Robust build infrastructure enabled rapid iteration"	Resilience through infrastructure	The investment in continuous integration systems acted as a physical traceability protocol, allowing for real-time goal validation.	Technical Debt (Mitigation)
"Nemesis system churn... we are progressively returning to the original focus"	Volatility of innovation requirements	Highly complex systems require exploration cycles that can derail the design without a stable design baseline.	Scope Instability
"Urorks like snowflakes, hideously unique, through cross-disciplinary collaboration"	High-fidelity asset synchronization	The technical compliance of procedural assets requires the deliberate breaking down of silos between art, storytelling, and AI engineering.	Fragmentation and Misalignment

5. Final considerations of the case

The analysis of Shadow of Mordor suggests that technical stability in complex systems is a product of focused governance. While most of the sample suffered from fragmentation, this case shows that resilience depends on the ability to align the studio to a single objective and sustain that vision with a robust software infrastructure. Ontological attrition was partially managed by transitioning from a prescribed multi-game task to a real activity centered on procedural innovation, indicating that traceability between requirements and execution is a decisive metric of technical compliance.

CASE A10 - TOEM (SOMETHING WE MADE)

1. Case identification

- **Project:**TOEM.
- **Studio:** Something We Made.
- **Central condition of the case:**Dissonance between design intent and public perception.
- **Documentary source:**Post-mortem of the studio technician and associated public discussion.

2. Engineering context and activity

Something We Made faced a critical software identity challenge during the development of TOEM. The engineering activity – understood as the effective result of execution under pressure from market feedback and genre uncertainty – revealed an ontological mismatch between the developers' intention and the end-user's perception. While the prescribed task was to create an adventure game, players who tested the demo on Android categorized it as a puzzle. The need to perform multiple reactive iterations after public exposure, to make the artifact match the original genre vision, reveals an Ontological Friction.

3. Extraction via GQM paradigm

Matrix A10.1 - Product message alignment

Goal	Question	Metric	Analysis/Response
To solidify the artifact as an adventure software, avoiding association with typical puzzle mechanics.	How did the user's differing perceptions impact the project's timeline and design architecture?	Reports of feedback via the Android demo indicated that multiple iterations were needed to change the direction of the game.	The failure to correctly externalize the design intent in the pre-production phase resulted in genre dissonance. Engineering operated under a fluid identity, generating refactoring efforts to decouple the game from unwanted puzzle conventions.
Technical Source URL: https://www.gamedeveloper.com/business/postmortem-something-we-made-s-i-toem-i- Reddit Source URL: https://www.reddit.com/r/IndieGaming/comments/pq0w8s/we_just_released_our_handdrawn_adventure_toem_ama/			

Matrix A10.2 - External Feedback Governance

Goal	Question	Metric	Analysis/Response
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Use industry showcases and playable demos as requirements validation protocols.	Did the delayed validation of requirements at public events expose weaknesses in early technical stabilization?	Identifying design churn after presentations at external events.	The reliance on external events to validate the basic premise indicates a pre-production flaw. The project moved forward without an internally validated experience baseline, turning public feedback into a trigger for late structural rework.
Technical Source URL: https://www.gamedeveloper.com/business/postmortem-something-we-made-s-i-toem-i- Reddit Source URL: https://www.reddit.com/r/IndieGaming/comments/pq0w8s/we_just_released_our_handdrawn_adventure_toem_ama/			

Matrix A10.3 - Stabilization of non-functional requirements

Goal	Question	Metric	Analysis/Response
Implement customized adventure systems that reject the standard difficulty scaling of puzzles.	How has the rejection of established mechanical standards increased the engineering burden?	The need for customized technical solutions instead of using native gender functions.	The technical decision to suppress gender stereotypes created a design vacuum that necessitated the development of unplanned proprietary mechanics, increasing complexity and technical debt.
Technical Source URL: https://www.gamedeveloper.com/business/postmortem-something-we-made-s-i-toem-i- Reddit Source URL: https://www.reddit.com/r/IndieGaming/comments/pq0w8s/we_just_released_our_handdrawn_adventure_toem_ama/			

4. Qualitative analysis framework - Gioia Method

Table A10.4 - Qualitative Analysis Structure of Case A10 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Players thought of the game as a puzzle... we wanted adventure."	Software identity dissonance	The conflict between the developer's mental model and actual usage reveals a breakdown in gender traceability.	Fragmentation and Misalignment
"We were forced into several iterations to change direction."	Fluctuation of requirements due to perception	Late validation forced structural refactoring to align the software with the original creative vision.	Scope Instability
"Feedback from GDC revealed that the message was not clear."	Disruption of intent traceability	The mismatch between product promise and technical output led to inefficiencies in production cycles.	Fragmentation and Misalignment
"We moved forward without a validated baseline of experience."	Lack of pre-production protocols	The shift towards production without internal entertainment benchmarks has transformed the audience into a tester of basic requirements.	Pre-Production Failure

5. Final considerations of the case

TOEM's analysis shows that Ontological Attrition does not stem solely from hardware or algorithm failures, but from a failure in the externalization of technical rationality. As a point of theoretical saturation, the case confirms that Fragmentation and Misalignment between what the software is and what it should be constitute the root cause of scope instability. Technical stability in projects of any scale depends on requirements governance that stabilizes the artifact's identity before the start of mass production.

CASE A11 - CELESTE (MADDY MAKES GAMES / EXOK)

1. Case identification

- **Project:** Celeste.
- **Studio:** Maddy Makes Games / EXOK.
- **Central condition of the case:** Extreme mechanical precision combined with clear emotional intent.
- **Documentary source:** Post-mortem of the studio technician and associated public discussion.

2. Engineering context and activity

Celeste's team sought to transform classic platforming mechanics into metaphors for Madeline's emotional journey in the face of anxiety. The prescribed task was to produce a platform game with extreme precision. The actual engineering activity, understood as the effective result of execution under the pressures of a highly difficult design, showed that, to sustain this level of precision without generating excessive frustration, the studio needed to develop real-time adjustment tools that facilitated iteration on the level design. Ontological friction was mitigated by a technical infrastructure geared towards reuse and rapid response to design adjustments.

3. Extraction via GQM paradigm

Matrix A11.1 - Feedback governance and mechanical sensitivity

Goal	Question	Metric	Analysis/Response
To synchronize extreme mechanical precision with the intended emotional response for the player.	How has fine-tuning micro-latency techniques, such as Coyote Time, mitigated the friction between prescribed difficulty and user frustration?	Technical implementation of tolerance frames, such as Coyote Time and Jump Buffering, documented in development reports.	The team used engineering to adjust the player's actual activity to the prescribed task of overcoming the challenge. Governance focused on the physical feel of the code transformed potential input failures into evidence of narrative compliance.

Technical Source URL:

<https://www.gamedeveloper.com/design/postmortem-maddy-makes-games-i-celeste-i->

Reddit Source URL:

https://www.reddit.com/r/celestegame/comments/823yye/celeste_ama_answers/

Matrix A11.2 - Iteration Scalability

Goal	Question	Metric	Analysis/Response
Reduce the response time between creative change and technical validation in the game engine.	How has investment in hot-swapping infrastructure prevented the accumulation of technical debt in highly complex sequences?	Using tools that allow editing the level design without resetting the software state at runtime.	The adoption of infrastructure for rapid iteration allowed the technical environment to act as an enabler of creativity. The case suggests that investing in tools from pre-production onwards reduced fragmentation and increased the ability to fine-tune the design.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-maddy-makes-games-i-celeste-i- Reddit Source URL: https://www.reddit.com/r/celestegame/comments/823yye/celeste_ama_answers/			

Matrix A11.3 - Conformity of experience and metaphor

Goal	Question	Metric	Analysis/Response
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To translate the protagonist's psychological state into the player's physical activity.	Did the search for the right emotional context for technical challenges compromise the early stabilization of requirements?	Reports of finding the right sentiment through exhaustive iteration in sequences without checkpoints.	Celeste's technical effectiveness lies in its ability to translate qualitative requirements into stable physical activity. Proactive governance prevented the volatility of requirements from turning into compliance collapse, preserving traceability between design and implementation.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-maddy-makes-games-i-celeste-i- Reddit Source URL: https://www.reddit.com/r/celestegame/comments/823yve/celeste_ama_answers/			

4. Qualitative analysis framework - Gioia Method

Table A11.4 - Qualitative Analysis Structure of Case A11 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Our task was to find out which emotional context worked."	Requirements validation through perception.	The reliance on finding sentiments indicates that the project progressed with unstable qualitative requirements.	Pre-Production Failure
"We developed hot-swapping tools for real-world adjustments."	Customized iteration infrastructure	Investing in proprietary tools is the governance strategy to mitigate ontological friction.	Technical Debt
"The mountain forces Madeline to confront her mental health."	User experience compliance	Technical effectiveness is validated by translating the player's psychological state into their physical activity.	Fragmentation and Misalignment

"Chapter 9 without checkpoints was a major technical challenge."	Extreme scope complexity	Long, unrestorable sequences test the limits of the software's physical and emotional integrity.	Scope Instability
"The code was designed to be reusable."	Engineering governance maturity	The logic of reuse prevents the game engine from becoming a barrier to the original creative intent.	Fragmentation and Misalignment
"Polishing like Coyote Time ensured the necessary precision."	Refinement of mechanical conformity	Frame rate adjustment externalizes technical knowledge to ensure traceability of the game's feel.	Technical Debt
"A small team focused on a unique interdisciplinary vision"	Convergence of creative and technical vision.	Breaking down silos ensures that the narrative dimension and the mechanics compose the same artifact.	Fragmentation and Misalignment

5. Final considerations of the case

Celeste's analysis suggests that fragmentation and misalignment are recurring risks, but it also shows a case where these risks were successfully managed. Instead of a complete lack of traceability, the team built a technical infrastructure tailored to the project's purpose, making the game engine effectively usable for the creative intent. The case indicates that resilience in complex independent games depends on governance that prioritizes rapid iteration tools from pre-production onwards.

CASE A12 - AGAINST THE STORM (EREMITE GAMES)

1. Case identification

- **Project:** Against the Storm.
- **Studio:** Eremite Games.

- **Central condition of the case:** A blend of city-building and roguelite, under significant uncertainty in estimation and execution.
- **Documentary source:** Technical presentation by GDC Vault.

2. Engineering context and activity

Eremite Games faced the challenge of developing software that integrated two distinct genres: city-building and roguelite. The prescribed task required delivering a project on an unprecedented scale for the studio, while the engineering activity – the actual result of execution under the pressures of mechanical innovation – revealed that the unpredictability of the run system and forest mechanics made scheduling extremely complex. Proactive governance based on benchmarking spreadsheets and the imposition of a 25% financial safety net to absorb unavoidable technical delays mitigated ontological attrition in this case.

3. Extraction via GQM paradigm

Matrix A12.1 - Evidence-based governance

Goal	Question	Metric	Analysis/Response
Synchronizing innovative mechanics with a predictable production schedule.	How did the use of benchmarking spreadsheets mitigate the uncertainty generated by mixing different genres?	Systematic use of external data to support estimates of time and effort.	The studio used data-driven governance to transform planning into a tool for reducing systemic risks, anchoring creative activity in physical evidence and reducing scope instability.
Technical Source URL: https://gdcvault.com/play/1034422/-Against-the-Storm Reddit Source URL: https://www.reddit.com/r/Games/comments/yha2es/against_the_storm_eremite_games_roguelite/			

Matrix A12.2 - Financial infrastructure management

Goal	Question	Metric	Analysis/Response
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To ensure the sustainability of development in the face of technical delays and market fluctuations.	How did the financial safety net influence the technical stability of the artifact?	Inclusion of a non-negotiable clause allowing for a 25% financial margin of error in contracts.	Investing in a margin of error acted as a mitigator of technical debt, allowing engineering to refine complex systems for over a year without compromising the physical survival of the team.
Technical Source URL: https://gdcvault.com/play/1034422/-Against-the-Storm Reddit Source URL: https://www.reddit.com/r/Games/comments/yha2es/against_the_storm_ermite_games_roguelite/			

Matrix A12.3 - External infrastructure resilience

Goal	Question	Metric	Analysis/Response
Maintaining technical integrity and team stability in the face of macroeconomic crises.	How did external financial vulnerabilities affect the physical capacity to execute the engineering project?	Technical impact resulting from the 10% cut in international transfers.	Technical stability proved dependent on the resilience of the external infrastructure; although the operational shock generated friction, prior defensive governance prevented the flow failure from turning into a compliance collapse.
Technical Source URL: https://gdcvault.com/play/1034422/-Against-the-Storm Reddit Source URL: https://www.reddit.com/r/Games/comments/yha2es/against_the_storm_ermite_games_roguelite/			

4. Qualitative analysis framework - Gioia Method

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
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"We create benchmarking spreadsheets for everything."	Evidence-based governance	The use of external data stabilizes the design baseline, reducing the volatility of requirements during production.	Scope Instability (Mitigation)
"25% safety net clause in contracts"	Financial infrastructure management	The financial margin for error allows for technical refinement without the pressure of the studio's immediate survival.	Technical Debt (Mitigation)
"The mix of city-building and roguelite made meeting deadlines difficult."	Hybrid gender innovation	The uncertainty inherent in mechanical innovation demands flexible governance infrastructures to maintain compliance.	Scope Instability
"Using Harvest Logic in the engine"	Engineering governance maturity	Making the infrastructure truly usable for the project's purpose avoids systemic ontological friction.	Fragmentation and Misalignment (Mitigation)
"We were caught off guard by a 10% cut in transfers."	External infrastructure vulnerability	External and unpredictable factors directly impact the physical capacity to keep the technical team active.	Technical Debt
"We accept that the estimates were wrong."	Activity and task synchronization	Explicitly acknowledging estimation errors is a governance strategy to avoid debt crystallization.	Technical Debt (Mitigation)
"Spreadsheets kept the team healthy and motivated."	Psychotechnical resilience of governance	Transparency and data tools reduce the stress of real-world activity under pressure and uncertainty.	Fragmentation and Misalignment (Mitigation)

5. Final considerations of the case

The analysis of Against the Storm demonstrates that technical stability does not depend solely on creative vision, but on the robustness of the infrastructure to support changes and uncertainties. The case illustrates the application of a data-driven management logic, allowing the studio to effectively align its production processes and tools with the project's objectives. The trail of evidence shows that, although the project faced Scope Instability, proactive governance, through benchmarking and safety nets, prevented operational failures from resulting in technical collapse at launch.

CASE A13 - XCOM: THE BUREAU (2K MARIN / IRRATIONAL AUSTRALIA)

1. Case identification

- **Project:** XCOM: The Bureau.
- **Studios:** 2K Marin / Irrational Australia.
- **Central condition of the case:** Gender instability, organizational fragmentation, and constant refactoring of core systems.
- **Documentary sources:** Technical post-mortem and associated public discussion.

2. Engineering context and activity

The project's genesis was marked by a strategic trade-off of resources: while 2K invested massive capital in BioShock, Irrational's Australian studio was tasked with reviving the XCOM franchise as a first-person shooter. The engineering process revealed a path marked by aborted concepts and long periods of stagnation. Ontological friction manifested itself in the difficulty of establishing the software's identity, resulting in a combination of conflicting requirements that compromised the final product's conformity.

3. Extraction via GQM paradigm

Matrix A13.1 - Resource Governance and Strategy

Goal	Question	Metric	Analysis/Response
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To revive the XCOM franchise as a tactical first-person shooter under capital and personnel constraints.	How did the strategic prioritization of another IP affect the technical stability and progress of the XCOM architecture?	Reports of stagnant progress and productive inertia during years of development.	Failures in resource governance suppressed technical advancement. Actual activity resulted in stagnation, where the absence of validated technical milestones prevented engineering from establishing a solid baseline, creating a development vacuum.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-2k-marin-s-i-xcom-the-bureau-i-. Reddit Source URL: https://www.reddit.com/r/Xcom/comments/1kx5v1/the_bureau_xcom_declassified_is_a_mess/.			

Matrix A13.2 - Team integration and communication

Goal	Question	Metric	Analysis/Response
Synchronize development between geographically distributed studios.	Did organizational fragmentation and a lack of a common vocabulary hinder the externalization of technical constraints?	Reports indicate that studios operate essentially separately and are unable to work in a unified manner.	Geographic isolation acted as a physical barrier to the flow of knowledge. Without cross-functional traceability protocols, teams became watertight units, resulting in team disintegration and the loss of shared technical rationale.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-2k-marin-s-i-xcom-the-bureau-i-. Reddit Source URL: https://www.reddit.com/r/Xcom/comments/1kx5v1/the_bureau_xcom_declassified_is_a_mess/.			

Matrix A13.3 - Systems Governance

Goal	Question	Metric	Analysis/Response
To implement a fluid tactical combat system that combined real-time and turn-based combat.	Have the constant changes to the gameplay architecture compromised the physical integrity of the code?	Implementation and subsequent abandonment of systems such as Time Units and radial menus with 18 simultaneous skills.	Scope instability generated massive technical debt. Constant refactoring of core systems turned the software into a technical patchwork, highlighting the collapse of structural conformity.
Technical Source URL: https://www.gamedeveloper.com/design/postmortem-2k-marin-s-i-xcom-the-bureau-i-. Reddit Source URL: https://www.reddit.com/r/Xcom/comments/1kx5v1/the_bureau_xcom_declassified_is_a_mess/.			

4. Qualitative analysis framework - Gioia Method

Table A13.4 - Qualitative Analysis Structure of Case A13 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"Progress on the project has stalled... we weren't contributing."	Stagnation of strategic scope	The absence of validated technical frameworks hindered the advancement of the architecture, resulting in stagnation.	Scope Instability
"Basically separate... we couldn't work together."	Geographic and organizational fragmentation	Geographic isolation hinders the flow of knowledge, turning teams into isolated units.	Fragmentation and Misalignment

"A tortuous path of aborted concepts"	Volatility of technical identity	The constant pivoting of gender prevents the trail of design evidence from stabilizing.	Scope Instability
"Implementation and subsequent abandonment of complex systems"	Accumulation of useless refactoring	Excessive rework on discarded requirements consumes the performance budget without generating value.	Technical Debt
"Technical leadership abandoned the project"	Erosion of governance rationality	The loss of specialized personnel disrupts technical memory, leading to flaws in the software logic.	Fragmentation and Misalignment

5. Final considerations of the case

The analysis of XCOM: The Bureau suggests that technical friction stems not only from hardware limitations but also from failures in externalizing governance rationale. The trail of evidence indicates that when technical leadership abandons a project amidst a lack of tangible progress, the software tends to become a combination of conflicting requirements. The case points out that technical stability depends on integrated requirements governance, and its absence compromises the final product conformity.

CASE A14 - ASTRONEER (SYSTEM ERA SOFTWARES)

1. Case identification

- **Project:**Astroneer.
- **Studio:**System Era Softworks.
- **Central condition of the case:**Late refactoring of core systems, rapid team growth, and tension between complexity and accessibility.
- **Documentary source:**Post-launch technical article and associated public discussion.

2. Engineering context and activity

System Era Softworks faced a critical challenge when preparing Astroneer's exit from Early Access: the crafting system was stagnant and perceived as shallow and repetitive. The actual engineering activity, understood as the effective result of execution under deadline pressures and design uncertainty, revealed a state of stagnation in creative actions. The team described the necessary refactoring as open-heart surgery on a conscious patient, indicating that core systems needed drastic changes while the software was already publicly operational. The association of conflict between divergent technical views on desired complexity and acceptable cognitive load for the player highlighted the Ontological Friction in this case.

3. Extraction via QQM paradigm

Matrix A14.1 - Design Architecture Governance

Goal	Question	Metric	Analysis/Response
Revitalize the crafting system to align gameplay incentives with the original design intent.	How have conflicting assumptions between engineering and design regarding cognitive load led to paralysis in development?	Reports of second-guessing within the team and the need for radical collaboration to challenge failed assumptions.	The operational delay reflected a form of technical opacity, in which the relationship between code changes and the final software behavior became unstable. Compliance was only achieved when governance allowed for the externalization of conflicting views to reduce the cognitive load on the system.
Technical source URL: https://www.gamedeveloper.com/game-platforms/3-years-of-astroneer-live-a-marketing-com Reddit source URL: https://www.reddit.com/r/Astroneer/comments/hf7sht/automation_update_patch_notes/			

Matrix A14.2 - Personnel governance and scaling

Goal	Question	Metric	Analysis/Response
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Maintaining technical cohesion and decision-making agility during the team's rapid growth.	How did the transition from a team of 5 to 40 members impact the traceability of engineering decisions?	Reports of growth from a minimal structure to a capacity of 40 people.	Sudden growth without a proportional adjustment in governance increased the risk of creating technical silos. The team needed to adopt new communication protocols to overcome fragmentation and maintain technical rationality regarding the software.
Technical source URL: https://www.gamedeveloper.com/game-platforms/3-years-of-astroneer-live-a-marketing-com Reddit source URL: https://www.reddit.com/r/Astroneer/comments/aj5zkg/lets_talk_about_dedicated_servers/			

Matrix A14.3 - Traceability of global accessibility requirements

Goal	Question	Metric	Analysis/Response
Minimizing language barriers through visual communication inherent in design.	Has the visual design strategy reduced the reliance on technical localization for entry into international markets?	Available in 13 languages, with almost 20% of revenue coming from the Asian market.	Requirements engineering anticipated the need for global software. Technical compliance was validated by commercial success in regions such as China, demonstrating that visually driven architectural governance acted as a mitigator of localization technical debt.
Technical source URL: https://www.gamedeveloper.com/game-platforms/3-years-of-astroneer-live-a-marketing-com Reddit source URL: https://www.reddit.com/r/Astroneer/comments/hhldie/thinking_about_getting_game_but_turn/			

4. Qualitative analysis framework - Gioia Method

Table A14.4 - Qualitative Analysis Structure of Case A14 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The crafting system was stagnant."	Erosion of technical rationality	The loss of control over the system's logic indicates a breakdown in the team's technical memory.	Fragmentation and Misalignment
"How to perform open-heart surgery on a conscious patient"	High-risk refactoring at runtime	The need to modify core systems while the software is in operation reveals a failure to stabilize requirements.	Technical Debt
"Fighting the problem of cognitive load"	Cognitive architecture conflict	The mismatch between the desired complexity and the required simplicity breaks the baseline of design.	Pre-Production Failure
"Mod support was never going to happen"	Strategic scope truncation	Suppressing promised features is a technical response to avoid maintenance collapse.	Scope Instability
"Growing from 5 people to 40"	Scaling of personnel governance	The team's growth requires new communication protocols to avoid technical silos.	Fragmentation and Misalignment

5. Final considerations of the case

Astroneer confirms that the project survived a state of Technical Opacity through a radical change in the governance of internal communication. Although it achieved significant commercial success, the trail of evidence shows that Ontological Attrition nearly paralyzed development at launch 1.0, due to poorly validated design assumptions. The case reinforces that even financially mature studios remain subject to fragmentation and the need for late technical interventions to recover artifact conformity.

CASE A15 - TURBO OVERKILL (SAM PREBBLE / APOGEE)

1. Case identification

- **Project:** Turbo Overkill.
- **Main author:** Sam Prebble.
- **Central condition of the case:** Development primarily focused on soil, with a high scope load and late mechanical refinement.
- **Documentary source:** Post-mortem technical analysis of the game and associated public discussion.

2. Engineering context and activity

Turbo Overkill was developed primarily by one person, who combined the roles of designer, programmer, and artist. The prescribed task was to deliver a fast-paced, visually high-fidelity FPS. However, the actual engineering activity, understood as the effective result of execution under the pressures of a solo developer and exhausting production cycles, revealed a massive overscoping issue: levels that previously took about a week to complete began requiring approximately a month of work in Episode 3, leading the developer to physical exhaustion.

3. Extraction via GQM paradigm

Matrix A15.1 - Validation of mechanical requirements

Goal	Question	Metric	Analysis/Response
Integrate vehicles, especially motorcycles, seamlessly into first-person gameplay.	Did the absence of early physical prototyping compromise the technical conformity of the vehicle system?	Reports of motorcycle mechanics being described as completely awful on the first try.	The failure to prototype the vehicle's physical interaction in the game engine revealed a mismatch between the prescribed task and actual feasibility. Compliance was only achieved through late reactive refactoring, highlighting a failure in the governance of mechanical requirements.

Technical source URL: <https://www.gamedeveloper.com/design/postmortem-turbo-overkill>
Reddit source URL: https://www.reddit.com/r/FPSGames/comments/16h5kcl/turbo_overkill_full_release_out_now_ama/

Matrix A15.2 - Production efficiency and sustainability

Goal	Question	Metric	Analysis/Response
Maintaining predictability in the delivery schedule within a stand-alone development model.	How has the increased complexity of the levels affected technical productivity and developer well-being?	A jump from one week to one month of work per level, and reports of physical exhaustion.	The exponential increase in effort required to maintain visual fidelity indicates an accumulation of operational technical debt. Solo governance has failed to mitigate overscoping, transforming engineering activity into an unsustainable process in the long term.
Technical source URL: https://www.gamedeveloper.com/design/postmortem-turbo-overkill Reddit source URL: https://www.reddit.com/r/FPSGames/comments/16h5kcl/turbo_overkill_full_release_out_now_ama/			

Matrix A15.3 - Review governance and knowledge silos

Goal	Question	Metric	Analysis/Response
Mitigate design flaws through technical leadership and external review.	Did the publisher's hands-free approach contribute to the crystallization of undetected design flaws?	Reports indicate that the publisher hasn't touched anything and there's a perception that	The absence of peer review or external technical leadership created a solitary decision-making silo. Without a trail of evidence of technical oversight, flaws in pace and density were only noticed belatedly

		the game is dragging on.	by the end user, disrupting the conformity of the experience.
Technical source URL: https://www.gamedeveloper.com/design/postmortem-turbo-overkill Reddit source URL: https://www.reddit.com/r/EPSGames/comments/16h5kcl/turbo_overkill_full_release_out_now_ama/			

4. Qualitative analysis framework - Gioia Method

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The motorcycle mechanics were completely awful at first."	Mechanical requirement dissonance	The failure to prototype physical interactions reveals the mismatch between the engine's intention and its actual feasibility.	Pre-Production Failure
"In the beginning it took a week per level; in the end, it took a month."	Productivity degradation due to complexity	The exponential increase in effort to maintain quality indicates an accumulation of operating debt.	Technical Debt
"Apogee didn't touch anything"	Centralization of knowledge in a silo.	The lack of external review masks design flaws until late stages of production.	Fragmentation and Misalignment
"Completely finished"	Inefficient personnel governance	Working under extreme pressure and without a support network leads to a collapse in the physical capacity to perform.	Fragmentation and Misalignment

"The game is dragging on."	Loss of rhythm consistency	The disconnect between developer ambition and user experience results in an excessively dense product.	Technical Debt
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5. Final considerations of the case

The Turbo Overkill analysis suggests that governance remains a determining factor regardless of team size. Although high technical capability was demonstrated in adjusting flawed mechanics throughout development, the project ended up reproducing patterns of scope instability and technical debt also observed in much larger productions. The case reinforces the thesis that Ontological Attrition stems from a lack of externalization and validation of requirements from pre-production onwards.

CASE A16 - STARFIELD (BETHESDA GAME STUDIOS)

1. Case identification

- **Project:** Starfield.
- **Studio:** Bethesda Game Studios.
- **Central condition of the case:** Massive ambition, legacy infrastructure, and fragmentation of the exploration experience.
- **Documentary source:** Institutional report on the behind-the-scenes aspects of the game, technical retrospective piece, and associated public discussion.

2. Engineering context and activity

Starfield represents Bethesda's most ambitious project in about 25 years, with an actual development spanning approximately eight years, beginning in 2015. The prescribed task was to deliver a continuous space exploration RPG with over 1,000 explorable planets. However, the actual engineering activity revealed a profound ontological conflict: the inability of the game engine to sustain the promise of an open world without fragmenting the experience into thousands of cells separated by loading screens. The difficulty in stabilizing the core loop in a timely manner, with the game only reaching a truly enjoyable form in 2022, the year originally planned for release, suggests the existence of Ontological Friction in the case of Starfield.

3. Extraction via GQM paradigm

Matrix A16.1 - Universe Simulation Governance

Goal	Question	Metric	Analysis/Response
To simulate a continuous and explorable universe that maintains Bethesda's open-world DNA.	Was the software infrastructure able to translate the scale of 1,000 planets into a smooth and uninterrupted browsing experience?	User reports indicate the screen is fragmented into small bubbles and loading screens are constantly appearing.	There was a breakdown in traceability between the creative requirement and the engine's physical capabilities. The instability of scope forced the adoption of a discontinuous exploration model, transforming the promised universe into a collage of procedural maps, negatively impacting immersion.
Technical source URL: https://news.microsoft.com/source/latam/features/vida-trabalho/nos-bastidores-de-starfield-onde-voce-pode-explorar-milhares-de-novos-mundos-entre-as-estrelas/?lang=pt-br			

Matrix A16.2 - Stabilization of fidelity and realism requirements

Goal	Question	Metric	Analysis/Response
Achieving a realistic level of detail based on technical documentation from NASA and SpaceX.	Did the exhaustive focus on astronomical detail and physics simulation compromise the early stabilization of gameplay?	An admission that the software only became fun seven years after development began.	The case reveals a pre-production failure in defining system priorities. Governance prioritized sensory fidelity and specialized knowledge over functional validation of the core loop, resulting in an excessively long and unstable development cycle.
Technical source URL: https://www.tecmundo.com.br/voxel/269053-historia-tras-starfield-jogo-demorou-25-anos-nascer.htm			

Matrix A16.3 - Human capital governance and specialization

Goal	Question	Metric	Analysis/Response
Integrate the team's diverse passions and technical knowledge into the artifact.	How has the management of highly specialized talent influenced software complexity?	Reports of developers integrating their side hobbies into the creation of characters and environments.	While the diversity of knowledge enriched the game's visual richness, it also exacerbated fragmentation. The attempt to cover as much as possible without a rigid traceability protocol resulted in an artifact where technical complexity became a barrier to agile iteration.
Reddit source URL: https://www.reddit.com/r/Starfield/comments/16gk750/is_it_open_world_or_not/			

4. Qualitative analysis framework - Gioia Method

Table A16.4 - Qualitative Analysis Structure of Case A16 (Gioia Method)

First-order concepts	Second-order themes	Justification of the path	Aggregate dimensions
"The most ambitious game we've ever built"	Ambition as a technical barrier	The excessive scale acts as a limiter of systemic integrity.	Scope Instability
"It frequently jumps from one map to another."	Fragmentation of spatial continuity	The infrastructure cannot support continuous charging, breaking the immersion.	Technical Debt
"It only really got fun in 2022"	Late stabilization of the core loop	The failure to validate fun and basic mechanics from the start indicates an ontological mismatch.	Pre-Production Failure

"We integrate hobbies like robotics and astronomy."	High-fidelity asset synchronization	Specialized knowledge without a stable baseline contributes to overscoping.	Fragmentation and Misalignment
"It's not an open world in the same way as Skyrim."	Gender expectation dissonance	The disconnect between marketing and actual activity reveals a governance failure.	Fragmentation and Misalignment

5. Final considerations of the case

Starfield's analysis confirms the invariance of governance in massive-scale projects. The case replicates patterns of scope instability and technical debt observed in other productions in the sample. The key difference here is showing that neither abundant financial resources nor extended deadlines mitigate ontological friction when the inherited infrastructure is unable to support creative ambition. The result was a fragmented experience, perceived as a partially unfulfilled promise.

Final considerations of the appendix

In summary, the cases analyzed demonstrate that the stability of digital game development processes depends on the early validation of assumptions, the alignment between organization and technical architecture, and continuous adaptation to the specific constraints of production. Thus, this appendix complements and supports the main discussion by consolidating the body of evidence that allows us to understand the mechanisms of conformity, deviation, and attrition throughout the projects.